



TEACHING NOTE

Clear Link Technologies, LLC: Driving Sales with Peer Effects

Facing declining sales growth, the executives at digital marketing and sales company Clear Link Technologies, LLC (referred to hereafter as Clearlink) decided to run a 4-week field experiment to test strategies for improving sales productivity. Although all work at Clearlink was performed individually, Clearlink executives hoped to leverage peer effects to generate knowledge spillovers between coworkers, boosting productivity. Designed in collaboration with a team of academic researchers, the experiment evaluates the effectiveness of various treatments to improve sales performance. The case examines the productivity impact of 1) group incentives, where two randomly paired coworkers are rewarded for joint output gains, 2) structured meetings, where two randomly paired coworkers participate in management-directed meetings about the sales process, and 3) a combination of the group incentives and structured meetings. These treatments are compared against an internal control group that was aware of the experiment, and an external group from a different office, that was not aware of any changes.

The case ends with protagonist Jessica Jones reviewing the data from the 4-week experimental period, wondering what conclusions she should draw from the results and what actions she should take next. The data show that all 3 treatments lifted concurrent sales productivity relative to both control groups. Students are left to debate the mechanisms driving the results and what the data reveals about the effectiveness of formal and informal incentives for fostering knowledge spillovers and peer effects at work.

In discussing the case, instructors will reveal -- often to students' surprise -- that Jones ultimately did nothing in the short-term, opting to wait for the longer-term results of the experiment before taking action. Instructors can then present students with the longer-term experimental results reported by Sandvik, Saoma, Seegert, and Stanton (2020). These long-run data show that group incentives resulted only in transitory sales gains, whereas treatments involving meetings between coworkers lifted sales for at least 6 months after the experiment ended. The reversion of gains under group incentives versus persistent long-run effects under the meetings treatment point to differences in the mechanisms at play behind each intervention. Specifically, group incentives led to a temporary increase in effort, while meetings created knowledge exchange between coworkers. The additional data allow students to

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refine their beliefs, assess and debate mechanisms, and form conclusions about the factors that block or enhance knowledge spillovers between peers at work.

The case serves four main purposes. First, it provides a new perspective on the importance of a) engineering peer effects and knowledge spillovers in the workplace, b) using effective knowledge management practices, and c) implementing novel practices for training and human capital development. Very little evidence exists on whether peer effects can be cultivated purposely, whether monetary incentives for group performance are effective for generating spillovers, or how process interventions may be required. Through the lens of experimental evidence, the case surfaces one of the only studies to date wherein a firm was able to engineer peer effects and observe the resulting outcomes. The case allows students to discuss the drivers and magnitude of spillovers between employees and how to create them. Since the main effects appear to stem from information exchange between coworkers, experienced executives will likely conclude that the firm's knowledge management practices were inadequate, and that lower-performers could have improved with better access to information about the best practices used by higher-performers.

The second purpose is to provide an example of experimental design in a challenging context. On the technical side, it allows students to explore how to design and implement business experiments in a real-world setting. For introductory classes, the case can be used to introduce fundamental concepts in planning and designing experimental tests such as control groups, randomization, and statistical power. For more advanced classes, instructors can focus on features that go beyond the textbook model, namely how caution must be exercised when experiments involve people to limit contamination between treatment and control groups.^a On the business side, the case teaches students to identify the organizational conditions that make experimentation useful. Overall, the discussion around experimental design allows students to build critical thinking skills for analyzing experimental findings while probing the strengths and weaknesses of various trial designs.

Third, the case prompts students to consider how the experimental results on the importance of knowledge-sharing can be translated into organizational initiatives. Students will brainstorm specific ways to adapt the experimental treatments to real-world settings. Through this process, they will also be introduced to aspects of organizational culture which can encourage or hinder knowledge-sharing in the workplace. In particular, the class discussion will review the importance of psychological safety, and three barriers to knowledge-sharing with peers: stigma, lack of visibility, and the adverse effects of incentives.

Finally, students will reflect on effective programs to augment and invest in workers' human capital development. Instructors can reveal that a follow-up experiment shows that many lower-performing workers fail to select into a similar mentoring scheme when it is a voluntary part of training, while a mandatory version of the program accelerated similarly positioned workers' career development. The discussion will challenge students to weigh the benefits of strategies that are paternalistic, where workers are required to take actions, against concerns about employee dissatisfaction.

The rest of this note provides (1) an overview of the learning objectives, (2) sample assignment questions, (3) a pedagogical overview, including case positioning, and (4) the class discussion plan. To

^a Although the case assumes no prior knowledge of statistics, it can also be used with students who are well-versed in research methods. Advanced courses may delve into nuance around running experiments inside firms or settings where experimentation requires care and thought around issues like Hawthorne effect, different treatments that participants might observe, and clustered inference problems.

encourage adoption in a wide range of management curricula, we suggest ideas for modifying the material based on the student audience, as well as modifications for teaching in an online environment.

Learning Objectives

- **Evaluate when to use experiments to inform business strategy and learn how to design meaningful experiments.** As firms increase their reliance on data to make decisions, executives should understand when existing (observational) data are reliable and when experiments are instead necessary for valid inference. By the end of class, students should be able to define basic experimental concepts, and to describe how they can be used to enhance the scientific rigor of an experiment. Classes with more experienced students will be challenged to engage more deeply on issues of experimental design and to suggest alternative design choices. Students will also consider how the results of one experiment might translate to other organizational settings and conditions.
- **Understand the causes and productivity dispersion across people.** There is significant work in economics that focuses on persistent performance differences across firms, but there is less appreciation of persistent performance differences across people. The case allows students to understand credibly estimated differences in performance across employees and to begin to appreciate that lower-performers' productivity may be lifted by effective knowledge management practices or trainings.
- **Evaluate barriers to knowledge-sharing and practices for stimulating peer effects.** Many students and experienced managers will likely recognize the importance of on-the-job learning that takes place via informal channels, but it is rarer for managers to consider whether intentional practices can accelerate learning and knowledge sharing among coworkers. The case allows students to see evidence-based examples of what works to facilitate knowledge sharing.
- **Consider how to blend formal and informal incentives.** In many firms, there is no monetary reward for individual tasks, which might lead to inadequate effort on unrewarded dimensions, like sharing information or helping others. At the same time, firms can use informal incentives (like promotions, non-monetary rewards, or even punishments after norm violations/relational contract violations) to coordinate behaviors. The case allows students to discuss whether formal incentives for cooperation are needed in an environment where employees actually have incentives, albeit weak ones, to compete with one another.

Suggested Assignment Questions

- 1) What is a peer effect? Have you experienced positive peer effects in your professional career? Did those experiences arise organically or intentionally? In particular, what types of environments or settings are most conducive for fostering positive peer effects, and how would you determine if your intuition is correct?
- 2) What is your assessment of productivity dispersion across people at Clearlink? What do you think drives this dispersion? How does productivity dispersion impact the firm's unit economics?
- 3) Case **Exhibit 8** provides average revenue-per-call by treatment group during the four weeks of active treatment and the four weeks preceding the experiment. What conclusions

would you draw from these data? What do you think is the mechanism underlying the differences across treatment? Is one treatment more effective than the others?

- 4) What should Jessica Jones do now? How should she incorporate learnings from the experiment through the rest of the organization to foster greater collaboration among sales agents? Would you expect the effects to change if Clearlink continued the treatments over the long-run?
- 5) Subjects of an experiment will sometimes alter their behavior by virtue of being observed. This is commonly referred to as the Hawthorne effect. Should Clearlink executives be concerned that the results are due to the Hawthorne effect?
- 6) Do you agree with the decision to test three active treatments; i.e., Pair-Incentives, Structured-Meetings, and Combined? Did the team fail to test other compelling interventions?

Supplemental Readings

The case themes touch on several topics that have been addressed in the literature on personnel and labor economics. The original papers can serve as helpful background for instructors, while this summary might spark directions for class discussions.

- The original research paper that inspired the case was coauthored by two of the case authors. This research paper provides more detail about the experiment's design and can be used to inform teaching in an advanced class. Please see Sandvik, Jason, Richard Saouma, Nathan Seegert, and Christopher Stanton (2020). "Workplace Knowledge Flows." *Quarterly Journal of Economics* 135(3): 1635-1680.
- A subsequent paper (Sandvik, Jason, Richard Saouma, Nathan Seegert, and Christopher Stanton, *Management Science*, Forthcoming) tests a follow-up question around whether the right workers will sort into workplace programs to develop human capital.¹ Instructors who want to frame questions around the role of active management or nudges to get workers to take beneficial behavior might benefit from familiarizing themselves with the results. The experiment described in the paper implemented a mentoring program using the Structured-Meetings treatment protocol (described in the case) for new hires at the firm. In the experiment, new hire cohorts were randomized into a "Voluntary" arm and a "Mandatory" arm. Those in the Voluntary arm were given the opportunity to opt into the mentorship program. Of those who opted in, mentorship was randomly allocated to some agents. Workers in the Mandatory arm did not have the choice to opt out of the program up front.

Those who received mentorship under the Mandatory arm had large gains from the program compared to non-mentored agents. However, those who received mentorship under the Voluntary arm had much lower treatment gains compared to those in the Mandatory arm. This result highlights the role of selection bias when mentorship was voluntary. When given the choice, lower-performing agents – those who would have derived the most benefit from mentorship – opted out of the program. In contrast, since lower-performing agents in the Mandatory arm received mentorship, the productivity gains in this arm were significantly higher.

- This result underscores a critical insight for firms: simply offering resources does not guarantee that they will be utilized by those who need them most. Firms must be proactive

in designing development programs in ways that address selection bias and ensure equitable access to opportunities for skill development. Firms that prioritize inclusive participation in such initiatives can unlock substantial productivity gains, especially by supporting employees with the greatest potential for improvement.

- This example also points to inference difficulties in experiments with selective recruitment. Effects estimated from pilot programs run on eager participants, even if there is randomization, may not generalize to a program at scale if the subject pool is not representative of the population.
- Instructors may wish to provide background reading to help students appreciate the difficulty of engineering peer effects in the workplace. An unsuccessful attempt to manufacture peer effects took place at the US Air Force Academy, an accessible synopsis of which can be found by Vedantam, Shankar. "Air Force Academy Squadrons Test Peer-Effect Assumptions." NPR, <https://www.npr.org/2014/03/26/294639911/air-force-academy-squadrons-test-peer-effect-assumptions>. The full article is from Carrell, Sacerdote, and West (*Econometrica*, 2013).² Another article surfacing the difficulty of engineering peer effects is Duhigg, Charles. "What Google Learned From Its Quest to Build the Perfect Team." *New York Times Magazine*, February 25, 2016.³
- To help explain the lack of knowledge-sharing among sales agents prior to the Clearlink experiment, instructors may wish to share resources describing why employees may avoid asking their peers for help. Even when doing so is in their best interest, employees may avoid asking for help if they fear that doing so will signal low ability to their peers (Chandrasekhar, Golub, and Yang, 2018)⁴ and managers (Heursen, Friess, and Chugunova, 2023).⁵
- Research conducted during the shift to remote work further captures that knowledge-sharing is adversely affected by physical distance (Emanuel, Harrington, and Pallais, 2023)⁶, and that remote communication limits employees' willingness to help one another (Roghanizad and Bohns, 2022).⁷ These barriers are further amplified when compensation structures induce competition between employees (Cornelissen, Dustmann, and Schönberg, 2023; Lazear, 1989).⁸
- Instructors might also share resources on building effective teams that encourage knowledge-sharing. HBS Professor Amy Edmondson, for example, stresses the importance of fostering "psychological safety" in the workplace, making employees feel comfortable to share their ideas and ask questions without fearing negative consequences.⁹ HBS Professor Jeff Polzer (with coauthors) offers strategies for preventing subgroups from forming within larger groups that restrict group cohesion. He recommends that managers consider the demographic and geographic makeup of potential team members, emphasize group performance and identity, and support full-team communication.¹⁰
- Resources on experimental best practices may also help students and instructors in technical classes. As a starting point, we recommend List, John A., Sally Sadoff, and Mathis Wagner, "So you want to run an experiment, now what? Some simple rules of thumb for optimal experimental design." *Experimental Economics* 14 (4) (2011): 439. Other recommended resources to illustrate the power of experiments for firm outcomes are:
 - Luca, Michael, and Max H. Bazerman. "The Power of Experiments: Decision Making in a Data-driven World." (Cambridge, MA: MIT Press, 2020).

- Blake, Thomas, Chris Nosko, and Steven Tadelis. "Consumer heterogeneity and paid search effectiveness: A large-scale field experiment." *Econometrica* 83 (1) (2015): 155-174.

Pedagogical Overview

This case provides an introduction to the economics of peer and knowledge spillovers in the workplace and the basics of experiments in organizations. Managers across an array of industries are increasingly looking to apply the scientific method to business problems by collecting and analyzing data and designing business strategy accordingly. As a result, this case may be applicable to courses that cover experimentation/data analytics, workforce management, training, sales, and general management.

At Harvard Business School, it has been taught in the second-year MBA elective course From Data to Decisions: The Role of Experiments, the second-year MBA elective course Managing the Future of Work, as well as in the school's executive education programs.

In the Harvard Business School Managing the Future of Work curriculum, Professor Stanton has paired the Clearlink case with material focused on remote work, alternative work arrangements, and barriers to knowledge-sharing as a lens to consider other real-world settings for workplace spillovers.

- 2 cases have historically been used in MFW that focus on the technology behind remote work and the organization of remote teams. These cases are:
 - Kominers, Scott Duke, Christopher Stanton, Andy Wu, and George Gonzalez, "Zoom Video Communications: Eric Yuan's Leadership During COVID-19," HBS No. 821-014 (Boston, MA: Harvard Business Publishing, 2020).
 - Stanton, Christopher and Shikhar Ghosh, "Collage.com: Scaling a Distributed Organization," HBS No. 817-038 (Boston, MA: Harvard Business Publishing, 2016).
- In addition, HBS Professor Tsedal Neeley's work, summarized in her book "Remote Work Revolution: Succeeding from Anywhere," (London, UK: Harper Business, 2021) can allow instructors to offer suggestions for effective management to facilitate effective communication and spillovers in remote teams.
- 3 cases introduce contingent work arrangements that may reduce the power of peer effects in traditional organizational arrangements.
 - Stanton, Christopher, Karim Lakhani, Jennifer Hoffman, Jin Paik, and Nina Cohodes, "Freelancer, Ltd.," HBS No. 820-075 (Boston, MA: Harvard Business Publishing, 2020).
 - Stanton, Christopher and Kristen Senz, "UST's Adoption of Open Talent," HBS No. 824-117 (Boston, MA: Harvard Business Publishing, 2023)
 - Kerr, William R. and Carl Kreitzberg, "Google: To TVC or Not to TVC?" HBS No. 820-048 (Boston, MA: Harvard Business Publishing, 2019)
- Finally, 2 cases have been used to illustrate barriers to knowledge-sharing in the workplace.
 - Kerr, William R., Federica Gabrieli, and Emer Moloney, "Transformation at ING (A): Agile," HBS No. 818-077 (Boston, MA: Harvard Business Publishing, 2018)

- Stanton, Christopher and Mel Martin, “Akooda: Charging Toward Operational Intelligence,” HBS No. 823-018 (Boston, MA: Harvard Business Publishing, 2023)

In the Zoom and Collage sessions, students debate whether the learnings might generalize to remote work settings and what practices need to be implemented so that remote workers are not isolated. Instructors who want to go down this path can reference Professor Neeley’s work along with some of the practice suggestions for remote team effectiveness in the following HBR piece by Sandvik, Saoma, Seegert, and Stanton (2020): <https://hbr.org/2020/11/research-how-virtual-teams-can-better-share-knowledge>.

In the Freelancer, UST, and Google TVC sessions, students are asked to consider how firm specific context, and the ability or need to learn from peers, affects the efficacy of arms-length gig talent compared to growing talent inside firms. This discussion relates to a broader point around the nature of the firm as an ecosystem, building off of seminal work by Alfred Marshall (1890) about how agglomeration economies arise through exposure to information or knowledge. The point of linkage from the case is that formal contracts did not facilitate this type of information exchange, and production processes based only on formal contracts will be unlikely to generate an environment where the whole is greater than the sum of its individual parts. In the context of Freelancer itself, this should help students to understand that gig talent models are most useful to firms when you can contract on outputs and for tasks where firm-specific, tacit knowledge requirements are minimal. Despite being able to contract on revenue at ClearLink, tacit knowledge is import for production, and the firm captures many of the gains from facilitating knowledge dissemination among the workforce.

The ING case traces the large bank’s introduction of agile work practices in their Netherlands home office. Part of the reason for the agile transformation is to encourage information sharing and to surface problems that workers face – allowing input from peers. There are many rituals and norms that agile teams use to share information and coordinate work, and these practices may form another example of ways to successfully engineer peer effects. The similarity with the Clear Link example is that workers reflect on where they are having problems and ask a group (ING) or a paired co-worker (Clear Link) for suggestions.

The Akooda case then raises the possibility that natural language technologies can be developed to provide context and knowledge about how others work, such that a future workplace that integrates similar technologies may be able to overcome the interpersonal frictions surrounding knowledge sharing more easily.

For instructors teaching courses on analytics or experimentation, the Clearlink case will likely come after a segment on A/B testing or on analysis of simple experimental designs. We hope the plans offered below will assist instructors in adapting instruction to suit their particular audience. To increase the accessibility of the case to a general management population, we also offer several supplementary readings that provide introductions to peer effects and to field experiments (refer to **Supplemental Readings**).

Discussion Plan

Starting with a high-level discussion about workplace peer effects, this teaching plan walks students through the process of running a business experiment, analyzing findings, and brainstorming and implementing organizational changes based on the results. Conducting the class in this order allows students to put themselves in the shoes of protagonist Jessica Jones as she considers how to stimulate

peer spillovers at Clearlink. Throughout the plan, we have also included adaptations for online teaching.

We suggest dividing an 80-minute class into the following five sections:

Topic	Time Allocation: General Management Setting	Time Allocation: Experiments or Other Technical Course
Introduction: Peer effects	10 minutes	5 minutes
Why use business experiments	10 minutes	5 minutes
Experimentation at Clearlink	10 minutes	25 minutes
Results	20 minutes	25 minutes
Promoting knowledge spillovers	15 minutes	10 minutes
Management strategies	15 minutes	10 minutes

Introduction: Peer Effects

Students have inevitably benefitted from peer effects at some point during their academic or professional careers, albeit they may need guidance to help recognize when. A provocative starting point might include: *“Imagine it’s Saturday morning, and you are about to leave your home for a long-planned day of relaxation with friends, when a colleague from work sends you an email asking you to come in to the office and help them with their work. The colleague is not your manager, nor do you manage them. How many of you would ignore the request?”* Ask students to raise their hands if they would ignore the request. It then helps to ask those who did not raise their hand why they felt drawn to help their colleagues. Almost any response will suffice (responses have generally included guilt, obligation, duty, and self-promotion), but it is essential to then ask the class as a whole: *“Do you think management intentionally engineered the organization to evoke such sentiments?”* The instructor could alternatively call on someone who had raised their hand earlier (i.e., they would opt to ignore their colleague’s request) to ask: *“What processes or organizational design decisions could management have implemented to at least give you pause before ignoring the email?”*

Students will often mention higher pay. If so, the instructor may want to ask, *“What kind of additional pay? Salary? Individual performance pay? Team performance pay?”* If students land on *team performance pay*, then it helps to probe why this would work; e.g., team pay would induce peer pressure to perform. The goal here is to establish that peer pressure through aligned incentives is one mechanism through which coworkers might influence individual output.

Once the idea has been fleshed out, the instructor may wish to write down the theory on the whiteboard (see Board 1 in **TN Exhibit 1**), before turning back to the class and asking: *“Moving beyond the Saturday email example, can you think of anything unrelated to pay that could foster peer effects in the workplace?”* The objective is to establish a distinction between using explicit incentives and more subtle human resource management techniques to foster workplace cooperation or knowledge spillovers. It is perfectly acceptable for the countervailing approach to differ from the *Structured Meetings* treatment described in the case, albeit a distinction is needed to motivate the next pasture on organizational experiments. Do not advance before recording a second, alternate theory on the board and the associated peer effect mechanism; e.g. *Having employees discuss work amongst themselves such that each can learn and incorporate the other’s best-practices*. This form of peer effects is often associated with imitation, wherein observing what coworkers are doing enables the dissemination of knowledges.

However, in situations where actions are shrouded or difficult to observe, management may need to encourage knowledge sharing between coworkers.

Instructors may frame the case discussion as analyzing the importance of these channels, and exploring related organizational changes that Clearlink could implement. Note, however, that peer pressure and imitation are not mutually exclusive. It is possible that incentive alignment may encourage knowledge sharing and peer pressure simultaneously. The experiment helps to uncover whether there is support for the perspective that the practices are complements or substitutes.

Why Use Business Experiments

Having established two sources of peer effects, the next objective is to motivate organizational experiments. This and the next discussion pasture make up the core of the class discussion for settings focused on experimentation. For class settings with less experienced participants, it can be helpful to review case **Appendix A** (“Background on Experimental Design”) to ensure that everyone has a similar comprehension level of basic experimental concepts.

A natural starting point to motivate experimentation is to ask: “How would we uncover whether one approach is more effective than the other, or if either is, in fact, effective at all?” Students who have read the case will immediately suggest experimentation, at which point the instructor can point out that experiments can be incredibly costly,^b and follow-up with, “What if the firm had previously used both the incentive scheme we have on one side of the board AND the implicit protocol on the other side of the board. Couldn’t we just look at performance during those periods?” To facilitate a robust conversation, ask probing questions such as, “How do managers make decisions?” “What is the business value of running experiments?” “Which are the best environments for conducting business experiments?” “What obstacles prevent companies from conducting experiments?” In doing so, try to elicit some of the following broad concepts and commentaries. It can be helpful to pre-board two column headers to guide the discussion: “Why experiment?” and “When to experiment?” (refer to Board 2 in **TN Exhibit 1**). Note: instructors who use this case in an experimental methods class or other technical course may want to spend less time on this section, or eliminate it altogether, due to students assumed prior knowledge. If you are teaching this case in an online setting, encourage students to respond to these questions using the chat feature.

Why experiment? Many management decisions are made based on intuition, anecdotal data, or correlational inferences. These sources of information are often incomplete or subject to bias. Experiments can bring scientific rigor to business, primarily through the use of randomized treatment and controls, which isolate the variable a researcher wishes to study. They can also help build support for new organizational practices among stakeholders. However, experiments come at the cost of attention and time, and there will be situations where running rigorous experiments may not be justified based on the benefit versus cost comparison.

When to experiment? Experimentation requires certain organizational conditions, resources, and expertise. For an experiment to produce useful results, the sample must be large enough to produce reliable inference. It is also helpful to use data to form baseline assumptions, and crucial to have a reliable way of collecting data. The performance metrics in question should be measurable and

^b Chapter 6 in Luca and Bazerman’s *The Power of Experiments: Decision Making in a Data-Driven World* (see **Suggested Readings**) describes an experiment in which researchers convinced eBay to temporarily stop paying Google for blanket search terms. It can be helpful to ask students to put themselves in the researcher’s shoes; namely: you are suggesting to the firm that the CMO-optimized, \$50 million annual spend on Google search terms might be for naught. Who would you need to convince to let you run such an experiment? Who might try to prevent you from running it? The takeaway here is that while experiments can be convincing, their adoption requires far more buy in than traditional (e.g., historical) analysis.

quantifiable. Importantly, organizations must be prepared to invest time and resources in executing the experiment (e.g., specific software tools, data collection and management). If the relevant expertise does not exist in the organization, it may be necessary to contract with an outside expert.

Another consideration is management buy-in and willingness to confront preexisting assumptions. Executives should be prepared to set aside their preconceived notions and be open to the results of the experiment. Some organizational cultures will be amenable to this approach, while others will find adoption difficult.

For executive audiences, the question of “why experiment?” may be relatively self-evident. They are likely to be more interested in discussing the conditions under which rigorous experimentation is appropriate. One key takeaway is that rigorous experimentation is a useful communication device to establish evidence in the face of skepticism. Most executive participants can recall a time in their career when they wished they could conduct a rigorous test to substantiate their position to colleagues, senior leaders, or other stakeholders. If you are teaching executives, we recommend spending the bulk of your time in this pasture on identifying—and setting—the appropriate organizational conditions for a business experiment with maximum impact. Push executive participants to share their experiences. Perhaps they used an experiment to persuade others, or perhaps they wish to design one that will add conviction to a decision they must make in the next few months. A useful line of questioning is to ask how an experimental approach compares to other forms of persuasion.^c How do they usually convince others to adopt a given strategy, and how might an experiment enhance their credibility in the boardroom? In our experience, many executives will report that they experiment, but do so using intuition rather than the rigorous planning embodied in the case.

We tend to end this line of inquiry by asking *which situations warrant a more rigorous, methodical approach?* We provide three key considerations:

First, rigor is useful to persuade stakeholders who suspect that a non-rigorous trial is only being conducted because the sponsor thinks the intervention will work. Such stakeholders may object to the results, saying “my situation is different.” The perception that tests occur where managers think an intervention will work often gives rise to this objection. A more rigorous approach can limit selection based on a gut feeling about what will work in a particular setting, allowing for the creation of generalizable insights.

Second, we recommend prioritizing rigor when decisions are difficult to reverse (e.g., situations with path dependence^d or sunk costs). These settings tend to tip the balance from speed to rigor.

Third, rigor is important when observational data are likely to be unreliable due to omitted variable bias, reverse causality, or other interpretation issues. A great example that we share with students is Blake, Nosko, and Tadelis’s (2015) experiment at eBay (as mentioned previously, this experiment is described in Chapter 6 of Luca and Bazerman’s book). In the experiment, eBay completely stopped bidding on its branded keywords in search engine advertising. Prior to the experiment, the executives at eBay had assumed that bidding on keywords was extraordinarily profitable, and a large share of their search engine marketing budget was spent on the “eBay” keyword. However, the experiment revealed that many of the clicks which had come through this advertising were actually from users

^c Ideas related to the interaction of persuasion, prior beliefs, and data are explored in Schwartzstein, Joshua, and Adi Sunderam (2024). “Sharing Models to Interpret Data,” (HBS Working Paper 25-011).

^d Path dependence refers to a phenomenon in which possible decisions in a given situation are shaped by prior decisions. Source: Oxford Reference, <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100310271>, accessed November 2020.

who would have navigated to eBay regardless. By rigorously testing the effectiveness of branded keyword spend, the experiment demonstrated that a large portion of the marketing budget was wasted, illustrating the pitfalls of relying solely on observational data to guide decisions.

Experimentation at Clearlink

The instructor should then transition the discussion to experimentation at Clearlink. We first state the problem facing Clearlink's leadership, and then ask whether rigorous experimentation would help solve this problem.

To define the problem, invite students to step into the shoes of Jessica Jones in the fall of 2017 as she prepares for a meeting about sales agents' performance. Remind students that the company is facing competitive pressure to improve its sales conversion rates (case p. 1). Among the various interventions under consideration is an effort to motivate salespeople to share their expertise with one another. Motivate this discussion by projecting case **Exhibit 3** for the class (or simply instructing them to refer to it in their version of the case) and orienting them to the way the data are organized. For many students, this data visualization will be unfamiliar; take a moment to explain the purpose of a histogram and define the axes.

As the data show, there is wide variability in sales productivity among sales agents. The company's leadership is curious to learn why some sales agents perform so much better than others. If Clearlink understood the reasons underlying outstanding performance, they might be able to elevate revenue-per-call (RPC, their targeted key performance indicator for the experiment). Referencing the chart, ask students a general question to invite brainstorming, such as, "*Seeing these plots, what would you want to know?*" or "*What might explain the variability in performance at Clearlink?*" There is no right answer; however, the following observations are likely to surface. The instructor may want to categorize these hypotheses into three categories: environmental conditions, innate ability, and acquired skill (see Board 3 in **TN Exhibit 1**). If teaching online, invite students to share their thoughts using the chat feature, and call on students whose written answers are particularly insightful.

- Tenure may improve agent performance (acquired skill). (Note: case p. 5 indicates that tenure was found to have a minimal effect on performance.)
- Some managers may be better than others, and can extract higher performance from their agents (environmental conditions).
- Certain products may sell more easily than others, advantaging some agents over others (environmental conditions).
- Agents may be subject to different incentive schemes (environmental conditions).
- There may be geographic or cultural factors that affect customer demographics and customer demand (environmental conditions).
- Perhaps seasonality plays a role (environmental conditions).
- Customer demographics could also differ by geography, increasing or decreasing the likelihood of making a sale (environmental conditions).
- A few people are performing a lot better than the average, achieving RPC between 150 and 200 during the four-week period. Perhaps these salespeople will always outperform the others (innate ability).

- Some agents may have learned to converse more with their (experienced) colleagues and in doing so, picked up tips for improving their performance (acquired skill).
- Some agents may be innately more talented at sales than their peers (innate ability).
- Perhaps the most talented sales agents have longer sales calls because they have learned that the longer a call lasts, the more likely it is that a sale will materialize (acquired skill).

The overarching question here is whether the dispersion in performance among agents is due to innate ability, skill, or luck/noise in the data. Articulate this dilemma for students by asking the following questions: “Are star sales agents born or made? Do all agents have equal access to success at Clearlink?” Students may turn to case facts to argue an intuitive position. For example, on p. 5, sales agents cited “‘superior ability to learn and respond to the customer’s needs,’ ‘a better sales process,’ ‘superior product knowledge’ and ‘superior effort,’” as causes for stellar performance. Other students may point to tenure on the job, a lucky string of calls, or other factors that may explain performance dispersion.

By allowing this discussion to go on, instructors can then make the point that this question is difficult to resolve given the data prior to the experiment. Pause here and note this difficulty. Instructors may also want to note that because of randomization, the experimental data in the case should allow students to conclude whether there is some ability to “make” a good sales agent given the right conditions.

To more explicitly motivate the experiment, ask students, “As you read in the case, Clearlink is looking to improve performance among its sales agents. Why should Jessica Jones use a business experiment to solve this issue?” Students are likely to point out that the organizational context benefits from measurable performance metrics (e.g., revenue-per-call (RPC)) that are “easily attributable to individual performance” (case p. 2). Clearlink also has a large enough employee population (730 sales representatives; case p. 2) to test different interventions. In addition, Clearlink has an established history of experimentation, suggesting that the organizational culture is open to this management approach (even if not all experiments to date have been scientifically rigorous; case p. 3). Overall, a rigorous experiment is a neutral mechanism for resolving Clearlink leadership’s internal debate about the drivers of productivity and knowledge transfer among sales agents, providing actionable information that could be applied across Clearlink’s sales teams.

Ultimately, Jones decided that experimentation would be beneficial. Instructors should pause here to review the key components of the Clearlink experiment. The objective of the experiment was to determine the best way to engineer peer effects among sales agents, and to test the effect of different practices on RPC. Sales agents in the experiment were divided into treatment groups: structured-meetings, pair-incentives, and combined (refer to case **Exhibits 4 and 5**). Those in the treatment groups and internal controls knew they were participating in the experiment. In contrast, the external control group was based at a separate office, and had no knowledge of the experiment.

Still, some students may remain unconvinced of the benefits of experimentation for Clearlink. Make space for skepticism by asking students, “Is anyone skeptical of the benefits of experimentation for Clearlink? If you were in Jones’s position, what approach might you use to resolve this issue?” As students respond, make note of their concerns on the board (refer to **TN Exhibit 1**). Students may caution that disseminating experimental findings across the organization has not been easy in the past (see Adam Mergist’s quote on case p. 3, which begins, “One team in our Orem office might be doing one thing, and thinks it’s

going great; meanwhile another team in the Salt Lake City office is trying another thing, which they think is going great....”).

In addition to broad concerns about experimentation, instructors should probe students to consider specific concerns about the experimental design. Instructors may ask questions such as “*Compared to other environments, what concerns would you have when designing an experiment at a call center like Clearlink?*”

One important difference is that the sales agents are aware that they are participating in an experiment. As noted on case p. 6, “Agents were informed that their productivity data would be studied by university researchers.” Students should bring up the potential for Hawthorne effect,^e the risk that experimental subjects may act differently under a researcher’s watch. Students may also point out that the potential for contamination—knowledge that different subjects have different treatment—could affect the results.

Another relevant concern relates to the length of the experiment. Some students may argue that a four-week experiment is not sufficient to understand whether learning or knowledge transfer took place. Students may express concern that by only studying these agents for a short period of time, the researchers may perceive novelty effects instead of real effects. Perhaps there is a ceiling on the usefulness of additional information in this context. Once agents learn a few tips from their peers, their learning may plateau.

Once students have offered a few critiques and concerns, discuss the ways in which the design the experimenters have chosen addresses and/or fails to address those concerns. For instance, to test for Hawthorne effects, the researchers chose to implement an external control group with no knowledge of the experiment and no connection to the other experimental groups. However, because the group is located at a separate site, it may have inherent differences (e.g., better managers, less stressful work environment) that could make it an imperfect comparison to the experimental subjects. To prevent contamination, the experimenters in this case used clustering, which helps to prevent spillovers between agents of different groups. (See below for an optional extended discussion of clustering.) As for the internal control, outcomes could be compromised by Hawthorne effects or contamination. Push on this last point by asking students, “*How do we figure out whether Hawthorne or contamination is occurring?*” To overcome these concerns, it is possible to compare the internal and external control group results with the pre-experimental period (readers with a background in data analysis will recognize that this is essentially a difference-in-differences empirical model).

Advanced analysis: clustering This optional section may be useful in settings focused on experimental design, and explores the design choice of clustering in greater detail. In the Clearlink experiment, sales agents are clustered into the same treatment by manager; hence, people who sit together get the same treatment, with identical conditions. There are benefits to this approach, including ease of implementation (“administrative convenience”) and preventing contamination between individuals. However, there are also costs to this design choice. For one, it requires a larger sample size to achieve the same statistical power as an individually randomized trial. In a clustered design, the threshold for minimum detectable effect size is higher (see case **Exhibit A1**) and depends on the intra-cluster correlation coefficient. How similar everyone is within a manager (cluster) determines the power of the cluster design. As a pre-design check, experimentalists will want to find out how correlated data are *within* versus *between* clusters. Ask students to consider if there are any additional experimental design choices that can solve this issue. For example, students may suggest

^e Hawthorne effect described atypical behavior among research participants who knew they were being studied.

dropping a treatment to increase sample size among the remaining treatment groups. Additionally, the researchers could choose to extend the duration of the experiment to reduce residual variability.

Results

Given the structure of the experiment and its associated benefits and concerns, invite students to predict which of the strategies—pair incentives or structured-meetings—will be most effective at driving sales performance. Call for one student to present each position. Refer to case p. 6, “Opposing Views,” for textual evidence. If you are teaching this case online, use the polling feature to launch a one-question poll asking: *Which will drive greater performance gains – pair incentives or structured meetings?*

Pair incentives Many believe that incentives will drive performance. In the past, Clearlink has employed and found success using both individual and team incentives. Money is inherently motivating, especially for those drawn to sales occupations. Money tied to joint output might lead to peer pressure, boosting effort, or to knowledge sharing and the dissemination of best practices.

Structured-meetings Those who advocate for structured-meetings generally believe that these management-driven sessions will remove the awkwardness of asking for help, and provide structure that facilitates a productive conversation. In this view, people are motivated by opportunities to socialize and share with their peers, especially when doing so might benefit them. This view further assumes that, prior to the experiment, agents were unable to imitate high-performing peers because of a lack of visibility into best practices, arising from barriers to asking for help. Participating in structured meetings would expose the most effective sales strategies, giving lower-performers the insight they need to imitate their high-performing peers.

After discussing students’ predictions, instructors should reveal information about the results of the experiment in stages. At each stage, students’ task is to put themselves in Jones’ shoes, determining whether they have sufficient information to act.

First, instructors should present the short-term results capturing the effectiveness of the treatments during the 4-week experimental period. According to case **Exhibit 8**, all three treatment groups saw a sales lift at the end of the experiment, though the structured-meetings and combined treatments seem to have outperformed the pair incentives. Given these results, ask students, “*What actions should Jessica Jones take?*”

If the goal is maximizing RPC, Jones might calculate the return on investment for each treatment before deciding what to implement over the long-term. To compare investments, consider the cost of providing incentives (\$40) to the cost of lunch (\$7) in the structured meetings. Based on these calculations, many students will likely recommend that Jones should redesign Clearlink’s ongoing training curriculum to emphasize peer relationships and mentoring. In pitching the changes to the CEO and her colleagues on the senior leadership team, students will probably note that Jones should stress the scientific rigor of the experiment and the potential to save on current incentive schemes that may not be as effective.

Students who are more skeptical may assert that differences between the groups are not large enough to draw a conclusion. Others may point out that the treatments have different opportunity costs and that Jones should consider, for example, whether increasing peer interactions could draw agents away from revenue-generating phone time. Others may suggest running the experiment again, perhaps at a different time of year, to try to replicate the results under different seasonal conditions, or to let the experiment continue running.

At this point, students will have only been presented the experiment results in a bar graph. Instructors might nudge students to consider whether certain information is obscured in this format. Namely, the bar graph presents only the average RPC across each period. For Jones to make an informed decision, she would likely want to observe changes in RPC during the experiment at a more granular level.

Instructors should end the discussion of short-term results with a poll: *What should Jones do next?* Where the options are “Implement the Pair Incentives Treatment”, “Implement the Structured Meetings Treatment”, “Implement the Combined Treatment”, and “Do Nothing.” It will not be obvious to most students, but we advocate that the right answer is for Jones to do nothing, at least for a period of time. The reason is that removing the active treatments will allow her to collect data on which treatments lead to persistent, long-term changes. The presence of long-term benefits could itself surface what went “right” during the study. In particular, the presence of long-term effects would suggest that the underlying mechanism is peer learning or knowledge acquisition, which occurred through the dissemination of best practices. This would be consistent with the imitation mechanism behind peer effects. On the other hand, if the experiment produced only short-term changes, then the likely conduit was peer pressure: agents were pressured into working harder, though not necessarily smarter.

At this point, instructors should project **TN Exhibit 2**, and ask students to describe the figure. *What timeframe does it capture? What do the axes represent?* Students should respond that the figure provides data for the seven weeks prior to the experiment, the 4-week experimental period, and the four-week period after the experiment ended (weeks 5+). The y axis represents RPC, and the information is presented separately for each of the 5 treatment groups.

Next, instructors should turn to describing the particular trends captured in the figure. *Focusing, for now, on the results before and during the experiment, what trends do you observe? Given the earlier bar graph, are you surprised by any of these results?* Prior to the experiment, RPC levels were relatively consistent across the 5 groups. As was captured in **Exhibit 8** of the case, all three treatment groups experienced an increase in RPC during the experimental period. However, the bar graph obscured how RPC differed week to week for each group. For the structured meetings and combined treatments, RPC shot up in week 1, and either increased or remained at a similar level up to week 4. In contrast, RPC for the pair incentives groups hit its highest point in week 1, dropped almost to pre-treatment levels in weeks 2 and 3, and then experienced a small increase again in week 4.

Then, shift students’ focus to the post-treatment period: *“What do you notice in the post-treatment data? What might explain the differences in performance outcomes across groups?”* The pair-incentives group quickly reverts back to the pre-treatment mean RPC, while sales gains persist in the structured meetings and combined groups. One explanation is that the pair incentives agents did not learn new techniques while paired with other sales agents, but instead increased their effort during the treatment period. Once the incentives were removed, their effort reverted to the pre-treatment level. Meanwhile, the structured-meetings and combined treatment groups learned new strategies that helped them sell more products. When sales agents confronted problems after the treatment period ended, they either drew on information learned from their peers during the treatment period or asked their peers for further assistance.

Emphasize to students that potential interpretations of these trends should explain both the overall results of the experiment, and the trends observed week-to-week. The effort mechanism in the pair incentives treatment would explain both of these trends; not only did effort decrease after the experiment, but it also decreased during the experiment itself. On interpretation is the following: during the experiment, the novelty of the incentives led to a substantial increase in effort in week 1.

However, since agents had not actually learned how to improve their sales strategies, that level of effort was unsustainable, and trended downwards, even as the experiment continued.

Instructors should now re-take the earlier poll; *Do you believe Jones now has enough information to make an actionable recommendation?* At this point, almost all students will likely agree that Jones has sufficient information. The instructor should agree with this decision, and display **TN Exhibit 3** as additional evidence that waiting to observe the long-term results was the correct choice. This exhibit captures the same information as the prior one, but now with data on an additional 16 weeks post-treatment. As expected, the post-treatment trends captured in **TN Exhibit 2** continue; RPC remains higher than pre-treatment levels for the meetings and combined treatments, while the pair-incentives groups' RPC looks indistinguishable from that of the controls.

Had the experimenters chosen to terminate the experiment and cease monitoring the agents after four weeks, the outcomes would have appeared vastly different. By continuing to monitor the subjects after the experimental period, Clearlink was able to identify the mechanisms underlying RPC shifts. Further, since the results indicate greater long-term benefits from the cheaper intervention (structured meetings), waiting for these results may have saved the firm costly investments which would not have led to long-term benefits.

Instructors should also encourage students to think about the best way for Jones to present these results to the rest of the leadership team. *How should Jones convey her confidence in the results and secure buy-in from her colleagues? How should she frame her "pitch" to ensure maximum impact?* Jones should emphasize the scientific rigor of the study to convince her colleagues of the meaningfulness of the results. Further, Jones should display the long-term results, and emphasize knowledge-sharing as the primary mechanism. She should make her recommendations practical and digestible; emphasizing the relatively low-cost of the meetings treatments, relative to their effects on productivity, may help her make a case for changing Clearlink's training and development programs. Specifically, given the long-term impact of the structured meetings, Jones may suggest that Clearlink implement regular meetings between peer sales agents. In the following discussion pasture, students will discuss specific strategies for designing and implementing these meetings.

How to Promote Knowledge Spillovers and Learning at Work

Overall, the experiment exemplified that productivity spillovers should be stimulated through imitation and learning, rather than peer pressure. In this pasture, students will explore how Clearlink might adapt the experimental meetings treatment to improve sales productivity in light of this finding.

As noted in the case, agents in the *structured-meetings* treatment completed worksheets which asked them to reflect on and share their experiences. Invite students to reflect on the worksheet used for meetings in the experiment. *Was the worksheet necessary to stimulate knowledge-sharing? What are the benefits and drawbacks of structuring a meeting in this way?*

The worksheets served several purposes in the meetings. First, they promoted self-reflection. Employees completed the front side of the form prior to the meeting, as shown in case **Exhibit 6**. This side asked them to recall their most and least successful calls of the past week. It also prompted them to set two goals for the rest of the week, implicitly promoting a growth mindset.^f By introducing this worksheet, management was signaling to employees their belief that sales skills can be learned through

^f Psychologist Carol Dweck popularized the term growth mindset to refer to a belief that one's abilities can be improved through effort. Source: "Dr. Dweck's research into growth mindset changed education forever," Mindset Works, <https://www.mindsetworks.com/science/>, accessed October 2020.

self-reflection and peer coaching. In effect, Clearlink's leaders were empowering their employees to take charge of their own job training and development. Once completed, the worksheets also served as a visual aid for the meeting, keeping agents on task as they discussed their recent performance. Without written responses to refer to, employees may not have known what to discuss and may not have been able to call to mind the highlights of their recent performance; the preparatory work written on the sheet could be used as a reminder. In principle, the worksheet also helped structure the conversation by narrowing the discussion focus, although conversations may have, in practice, ventured outside of this structure—either veering off course or interrogating issues of sales performance in greater detail.

The second side of the sheet (see case **Exhibit 7**) contained a questionnaire intended to facilitate dialogue. The questions mapped to the questions on the front side. As with the front side, there was room on the sheet to record responses. The questions required each member of the pair to listen to their peer. As the instructions dictated, "Please TALK to your partner about the questions below and write down their responses." Instructors should note that, although it was optional, about 80% of agents in the structured-meetings condition did complete a worksheet during each week of the experiment.

As for the drawbacks to using the worksheet, students may mention that the worksheets' structured nature could have inadvertently created a transactional dynamic, where employees focused more on completing the form than engaging in genuine dialogue. This could limit the potential for fostering deeper social bonds, which are often critical for long-term knowledge sharing. Additionally, the rigidity of the structure may stifle organic conversations that could lead to unexpected insights or creative problem-solving. Some employees might also feel that the structured format implies a lack of trust in their ability to navigate discussions independently, potentially dampening their enthusiasm for the process. Lastly, the incentive of receiving a free lunch for participating in meetings may have encouraged some agents to "game the system," prioritizing surface-level engagement with the worksheet and discussion over genuine effort to improve or share knowledge.

At some point during this discussion, a student is likely to interject that the experiment cannot identify knowledge spillovers from peers from the self-reflective prompts and other factors in the worksheets. If this comes up, acknowledge that the comment is astute, while displaying **TN Exhibit 4**. One way of disentangling self-reflection from knowledge spillovers is through peer quality. If gains were only due to self-reflection, one would not expect that the identity of a peer would matter for RPC. If instead, knowledge flows explain the gains, then having a better peer should be associated with greater RPC gains. **TN Exhibit 4** shows the effects of having a better, randomly selected peer partner on RPC are positive – and, in fact, a pairing of high-performing and low-performing agents yields the biggest gains.

Imagine, now, that Clearlink decides to implement regular meetings between peer sales agents. Ask students to discuss how they'd want to adapt the original worksheet, or whether they'd want to utilize a worksheet at all. After hearing some responses, inform students that a follow-up experiment was, in fact, conducted at Clearlink which explored how the organization should integrate knowledge-sharing into their employee development and training.

Specifically, the follow-up experiment investigated whether Clearlink's mentorship program should be voluntary or mandatory. In the study, new hire cohorts were randomly assigned to either a "Voluntary" group or a "Mandatory" group. In the Voluntary group, employees could choose to opt out of a mentorship program. Of agents who opted in, some were randomly selected to receive mentorship. In the Mandatory group, all agents received mentorship. Instructors may wish to project **TN Exhibit 5** to illustrate agents' allocation to groups.

Instructors may wish to pause here and ask students to predict what the results of the experiment were. *Do you think there were differences in productivity gains across the two groups? What mechanisms might be underlying those differences?* Some students might argue that forcing participation could reduce engagement or motivation, limiting the program's effectiveness in the Mandatory group. Allowing agents to opt in could ensure that the agents who do receive mentorship are those who will be the most engaged. Others might hypothesize that productivity gains might not differ significantly between groups, as the success of mentorship could depend more on the quality of the mentor-mentee match rather than whether participation was voluntary or mandatory. After hearing some responses, the instructor can then reveal the results.

The results showed that employees in the Mandatory group experienced significantly larger productivity gains than those in the Voluntary group. The authors explain the gap through selection bias. When participation was voluntary, lower-performing employees—the ones most likely to benefit—tended to opt out. Mentored agents in the voluntary condition did experience performance gains, but the shift was less significant than in the Mandatory group.¹¹

This finding emphasizes that voluntary participation can unintentionally limit the effectiveness of workplace programs by excluding employees who would benefit most from them. Firms aiming to foster widespread skill development and productivity improvements must design programs that proactively address selection bias. By requiring participation or offering incentives that encourage inclusion, organizations can ensure that all employees, especially lower performers, have access to opportunities for growth. However, making programs mandatory may decrease enthusiasm and engagement, as employees may feel coerced rather than motivated to participate. Striking a balance between inclusion and employee autonomy is essential to fostering both productivity gains and long-term buy-in.

Management Strategies

In this final pasture, instructors should guide the class in exploring alternative organizational practices that Clearlink could adopt. While the study focused specifically on the impact of worksheet-aided peer meetings, its findings underscore the broader value of peer knowledge-sharing as a driver of sales productivity. Although the changes Clearlink implements should aim to facilitate knowledge-sharing, they need not replicate the exact methods used in the experiment.

Instructors should encourage students to brainstorm alternative methods Clearlink could use to promote peer learning, keeping feasibility in mind (see Board 4 in **TN Exhibit 1**). For instance, agents might meet in larger groups instead of pairs. Alternatively, knowledge-sharing could be integrated seamlessly into agents' workflows, eliminating the need for dedicated meeting times. Another possibility could be matching agents based on performance metrics rather than assigning them at random. In the original experiment, Sandvik et al. (2020) found through an analysis of meeting worksheets that only 74% of lower-performing agents shared knowledge, compared to 93% of high-performing agents (p. 1672). Based on this insight, grouping high-performing agents with multiple lower-performing agents might increase the number of employees with access to expertise. However, such an approach would need to be carefully managed to avoid overburdening high performers or discouraging mid- and lower-performing agents. Additionally, shifting to a group structure might detract from the sense of genuine social connection that one-on-one meetings can foster.

As students discuss, be sure to highlight the core question: *What conditions are essential to foster meaningful learning between peers?* For example, for meetings to facilitate self-reflection and peer learning, employees must feel comfortable sharing their honest experiences—both successes and challenges—with their peers. This peer could be a trusted colleague, or it could be someone they have

little prior connection with. This comfort aligns with Amy Edmondson's concept of psychological safety: "a shared belief held by members of a team that the team is safe for interpersonal risk-taking."⁸ *What steps has Clearlink already taken to cultivate this shared belief? What additional practices could Clearlink implement to further enhance psychological safety and improve the effectiveness of peer meetings?*

Next, ask students to share their own experiences of self-reflection and learning at work. Perhaps students have worked in environments that emphasize agile design and development processes; these students may have attended daily stand-up meetings in which they reflect on the successes and failures of the previous day and coordinate their work going forward. Ask students how these meetings have been structured. *Is someone recording what is discussed, as in the Clearlink worksheets? How do the managers facilitate psychological safety?*

Instructors should then transition to broader questions about the culture at Clearlink. As the case describes, sales agents at Clearlink were aware of each other's performance prior to the experiment -- "Team members' average RPC figures were posted publicly on the sales floor, and company surveys showed that agents were keenly aware of (1) their relative standing and (2) the identity of the firm's top sales agents" (case p. 5). Since Clearlink agents earn commission pay, it is in their best interest to seek out opportunities to increase their sales productivity. However, the results of the experiment reveal that agents had not been utilizing their peers as resources to improve prior to the experiment. *What, then, prevented sales agents at Clearlink from asking for advice from their peers?* Instructors should present students with the following mechanisms that Hoffman and Stanton (2024) identify as limiting knowledge spillovers in organizations¹² (see Board 5 in **TN Exhibit 1**):

Advice-seeking stigma Workers may avoid seeking help due to perceived social or reputational costs. Chandrasekhar et al. (2018) demonstrate that individuals are 55% less likely to seek advice that could improve task outcomes when doing so signals lower ability. Their experiment found that signaling concerns, rather than feelings of shame, were the primary deterrent to advice-seeking.¹³ Similarly, Heursen et al. (2023) show that professionals reduce advice-seeking by 16% when the action is visible to a manager, despite managers' competence assessments being unaffected.¹⁴

Limited observability and distance Limited observability and physical distance can impede knowledge spillovers. Emanuel et al. (2023) find that software engineers in a Fortune 500 company received 22% more feedback on their code when co-located with their teams, compared to teams split across buildings. However, this advantage disappeared with the shift to remote work, highlighting the importance of proximity in fostering collaboration.¹⁵ Similarly, Roghanizad and Bohns (2022) demonstrate that help is significantly more likely to materialize when requested face-to-face, rather than through remote channels, like Zoom, phone, or email.¹⁶

Adverse effects of incentives Incentives tied to competition can hinder knowledge spillovers by creating conflicts of interest among peers. Cornelissen et al. (2023) study workers who have gone through different forms of training in the German labor market. They find that while higher-quality trained peers boost untrained workers' wages by 3.4% over five years, while the presence of high-quality untrained peers—who are seen as potential competitors—has a negative wage effect. These results suggest that rank-order tournaments for promotions may discourage cooperation and limit opportunities for knowledge-sharing.¹⁷ Whether competition suppresses knowledge spillovers entirely

⁸ Amy Edmondson, "Psychological Safety and Learning Behavior in Work Teams," *Administrative Science Quarterly* 44 (2) (June 1999): 350-383, <https://tinyurl.com/ycc5gupt>, accessed October 2018.

or merely obscures their effects remains unclear, as rivalry may lead to reduced collaboration or even sabotage (Lazear, 1989).¹⁸

Next, ask students to reflect on the role of these barriers within Clearlink's case. *Which of these factors do you think contributed to Clearlink agents' lack of knowledge-sharing with peers prior to the experiment?* Since agents worked in person and teams were located within the same office, advice-seeking stigma and adverse effects of incentives were likely the primary factors. An important point to bring up in this discussion is the fact that helping other agents could, in theory, reduce one's own compensation (case p. 5). While the case mentions that, in practice, commission changes are unlikely, *do you think this compensation structure may have nevertheless contributed to the lack of knowledge-sharing?*

Next, instruct students to re-evaluate their suggestions for practice changes from the previous pasture; *Do you think the solutions that have been proposed for Clearlink moving forward will be sufficient in helping agents overcome these barriers?*

Throughout this discussion, students will have likely suggested a variety of organizational changes that could potentially lead agents to feel overly observed. For example, requiring agents to attend mandatory meetings, regardless of their perceived usefulness, could feel overbearing, especially for high-performing agents. Similarly, implementing strict knowledge-sharing documentation or requiring managers to closely observe high performers might make agents feel micromanaged. While the results of the experiment point to the value of structured interventions, it's important to weigh that value against the risk of causing employees to feel overly monitored. Employees who perceive these practices as intrusive may respond by gaming the system—completing worksheets or attending meetings in a superficial way without engaging meaningfully. Furthermore, high-performing agents may feel singled out or overly scrutinized, which could lead to disengagement or even attrition, as they might prefer an environment that values autonomy over enforced collaboration.

The potential downsides of such overbearing practices highlight the need for thoughtful implementation. One way to address this is by emphasizing flexibility and personalization. For instance, rather than requiring all agents to attend the same meetings, managers could offer tailored workshops based on agents' self-identified areas for improvement. Additionally, building psychological safety is essential to counteract the negative effects of perceived surveillance. Managers can achieve this by framing interventions as opportunities for growth and development, rather than tools for oversight or control. For example, feedback should focus on celebrating progress and fostering collaboration, not on evaluating individual shortcomings. Encouraging agents to co-create their own development plans, rather than mandating participation in predetermined programs, can further ensure buy-in and alignment with individual goals. By striking this balance, Clearlink can maintain the benefits of structured knowledge-sharing practices while minimizing the risks of disengagement.

To conclude the session, ask students to return to brainstorming knowledge-sharing initiatives at Clearlink. *What are some specific ways managers could reframe these initiatives to feel less like micromanagement and more like empowerment?* Throughout this discussion, students should think through each initiative from the perspective of management, high-performers, and low-performers. *How might the risks of disengagement differ for low- versus high- performers under these interventions? If you were designing a knowledge-sharing initiative, how would you ensure buy-in from both high- and low-performers?*

Concluding Remarks

In concluding the discussion, ask students to reflect on general lessons they can take away from the experiment. We focus on the following four:

- Experimentation is costly, but can surface non-obvious conclusions and insights through careful design. Experimenting is useful when observational data are unreliable for inference or when rigor is needed to convince stakeholders to change their position.
- Management practice changes can influence peer effects in the workplace through two different channels. Incentive alignment stimulates peer pressure. Reducing barriers to coworker interactions stimulates imitation or the dissemination of best practices. Practices that facilitate imitation are likely to have longer-term benefits, and are especially relevant in environments like call centers where it is hard to observe what coworkers are doing.
- The importance of communicating with peers raises questions about distributed or remote work arrangements. What will generalize to digital environments and what types of peer interactions are more beneficial in person? It is possible that digital tools may ease coordination difficulties for managers to schedule structured meetings, but lack of face-to-face interactions may hinder the ability to imitate what others are doing.^h
- Finally, organizational changes to promote knowledge-sharing are likely to be the most effective when they contribute to building open and collaborative team environments. However, when designing initiatives that promote knowledge-sharing in hands-on ways, managers should ensure that employees do not feel overburdened or overly observed.

^h For additional discussion of these points, see Jason Sandvik, Richard Saouma, Nathan Seegert, and Christopher Stanton, "Research: How Virtual Teams Can Better Share Knowledge," *Harvard Business Review Digital*, November 2020, <https://hbr.org/2020/11/research-how-virtual-teams-can-better-share-knowledge>, accessed November 2020.

TN Exhibit 1 Board Plan

Board 1: Sources of Peer Effects	
Peer Pressure	Imitation
Requires goal alignment?	Can be based on observation or more direct knowledge diffusion
Less relevant in competitive environments?	Knowledge sharing required when observation is not possible

Board 2: Why Experiment?	
Why Experiment	When to Experiment
Scientific rigor	Strong performance metrics
Convince stakeholders	Sample size
Enhance data	Time/resources/expertise

Board 3: What Explains Performance Variability at Clearlink		
Environmental Conditions	Innate Ability	Acquired Skill

Source: Casewriters.

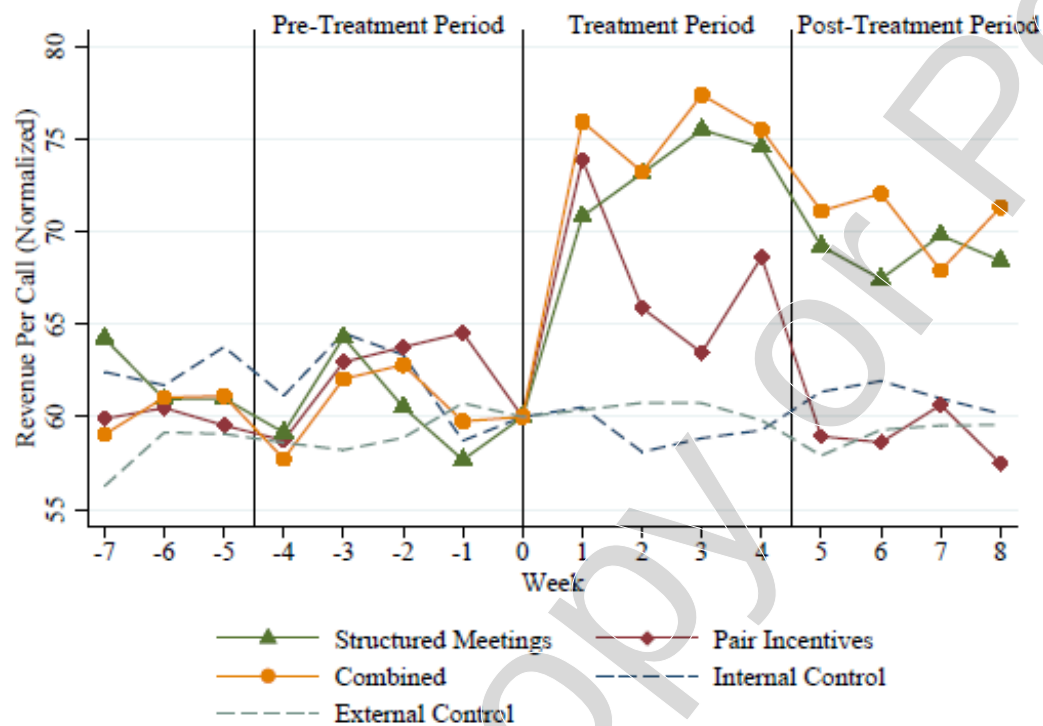
TN Exhibit 1 Board Plan (Continued)

Board 4: Possible Solutions		
Idea	Reduces Barriers to Knowledge-Sharing?	Avoids Employees Feeling Overly Monitored?

Board 5: Mechanisms Preventing Help-Seeking			
Psychological Safety	Help-Seeking Stigma	Limited Observability	Adverse Incentives

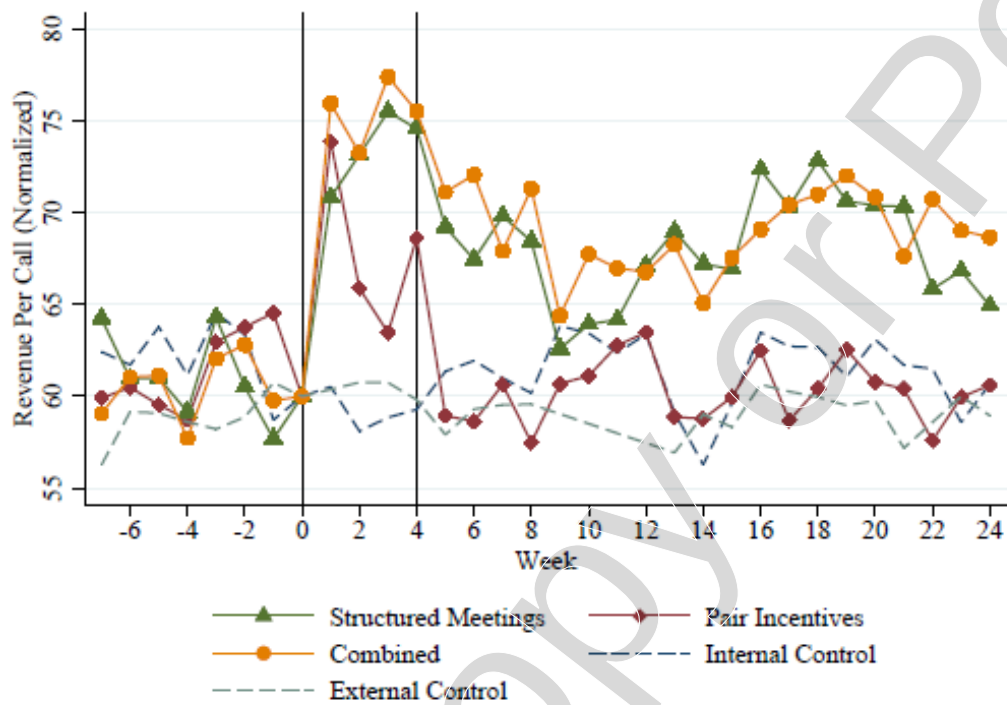
Source: Casewriters.

TN Exhibit 2 Medium-Term Productivity by Treatment

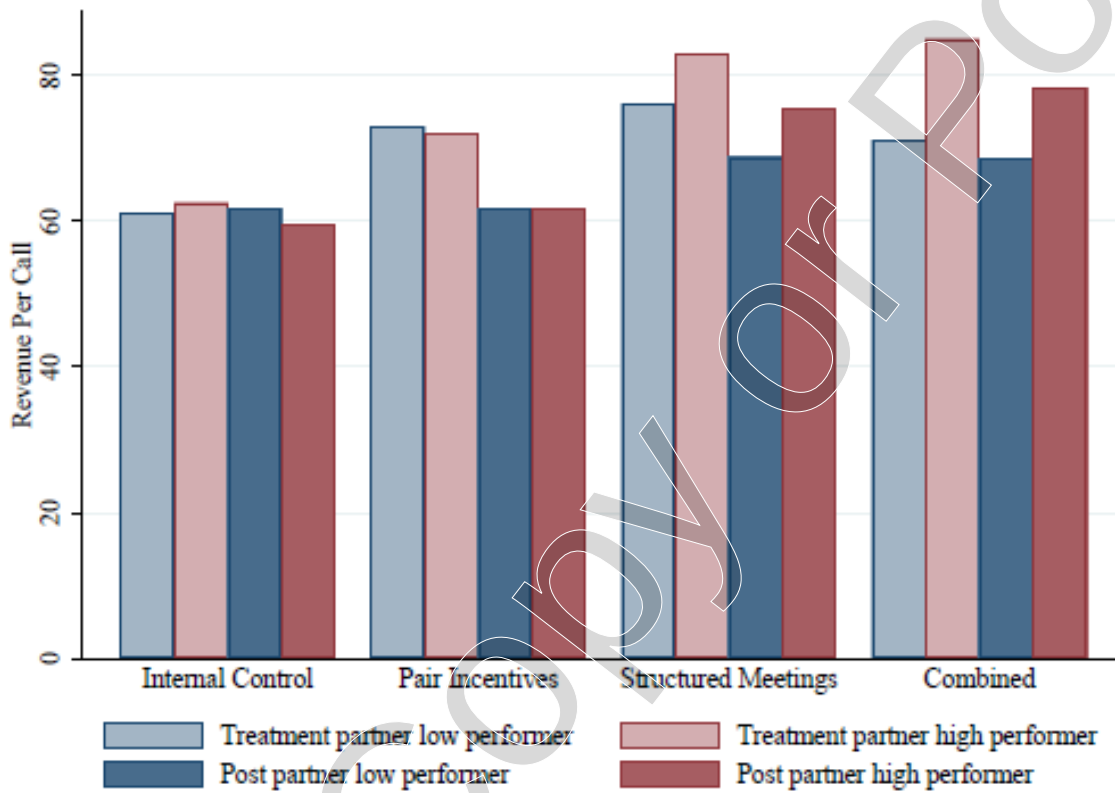


Source: Jason Sandvik, Richard Saouma, Nathan Seegert, and Christopher Stanton, "Workplace Knowledge Flows," *Quarterly Journal of Economics* 135(3): 1635-1680.

TN Exhibit 3 Longer-Term Productivity by Treatment

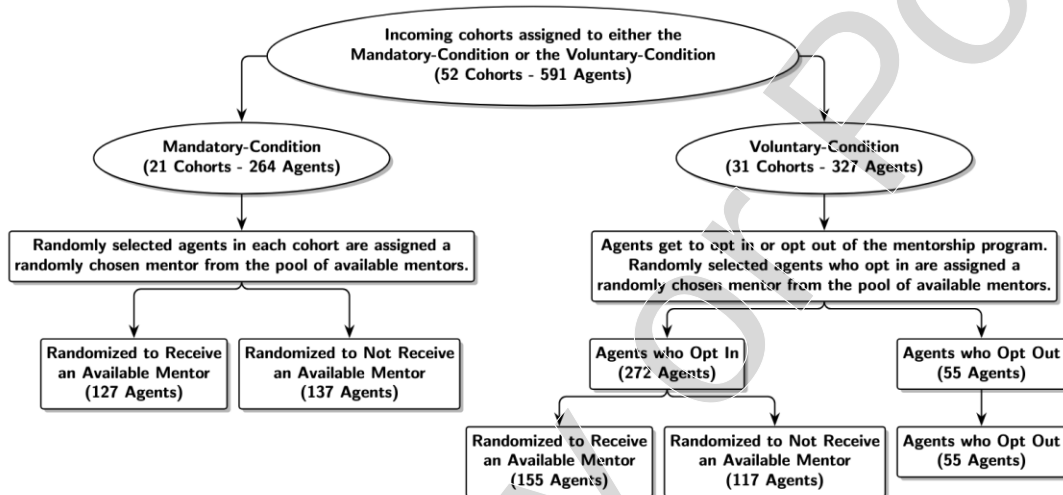


Source: Jason Sandvik, Richard Saouma, Nathan Seegert, and Christopher Stanton, "Workplace Knowledge Flows," *Quarterly Journal of Economics* 135(3): 1635-1680.

TN Exhibit 4 RPC Gains by Partner Quality in the Structured Meetings Treatment

Source: Jason Sandvik, Richard Saouma, Nathan Seegert, and Christopher Stanton, "Workplace Knowledge Flows," *Quarterly Journal of Economics* 135(3): 1635-1680.

TN Exhibit 5 The Design of a Follow-Up Experiment on Mentoring



Source: Jason Sandvik, Richard Saouma, Nathan Seegert, and Christopher Stanton. "Should Human Capital Development Programs be Voluntary or Mandatory? Evidence from a Field Experiment on Mentorship." *NBER Working Paper*: 29148, 2021.

Endnotes

- ¹ See Sandvik, Jason, Richard Saouma, Nathan Seegert, and Christopher Stanton (Forthcoming). "Should Human Capital Development Programs be Voluntary or Mandatory? Evidence from a Field Experiment on Mentorship." *Management Science*.
- ² Carrell, Scott E., Bruce I Sacerdote, and James West (2013). "From Natural Variation to Optimal Policy? The Importance of Endogenous Peer Group Formation." *Econometrica* 81, no. 3: 855-82.
- ³ See <https://www.nytimes.com/2016/02/28/magazine/what-google-learned-from-its-quest-to-build-the-perfect-team.html>
- ⁴ Chandrasekhar, Arun G., Benjamin Golub, and He Yang, (2018). "Signaling, Shame, and Silence in Social Learning." *NBER Working Paper*: 25169.
- ⁵ Heursen, Friess, and Chugunova (2023). "Reputational Concerns and Advice-Seeking at Work." Discussion Paper.
- ⁶ Emanuel, Natalia, Emma Harrington, and Amanda Pallais (2023). "The Power of Proximity to Coworkers: Training for Tomorrow or Productivity Today?" *NBER Working Paper*: 31880.
- ⁷ Roghanizad, M. Mahdi, and Vanessa K. Bohns (2022). "Should I Ask Over Zoom, Phone, Email or In-Person? Communication Channel and Predicted Versus Actual Compliance." *Social Psychology and Personality Science* 13, no. 7: 1163-1172.
- ⁸ Cornelissen, Thomas, Christian Dustmann, and UTA Schönberg (2023). "Knowledge Spillovers, Competition, and Individual Careers." Mimeo, and Lazear, Edward P (1989). "Pay Equality and Industrial Politics." *Journal of Political Economy* 97, no. 3: 561-580.
- ⁹ See "Psychological Safety and Learning Behavior in Work Teams," *Administrative Science Quarterly* 44 (2) (June 1999): 350-383, <https://tinyurl.com/ycc5gupt>, accessed October 2018.
- ¹⁰ See Polzer, Jeffrey T, C. Brad Crisp, Sirkka L Jarvenpaa, and Jerry W Kim (2006). "Extending the Faultline Model to Geographically Dispersed Teams: How Colocated Subgroups Can Impair Group Functioning." *Academy of Management Journal* 49 (4): 679-92.
- ¹¹ Sandvik, Jason, Richard Saouma, Nathan Seegert, and Christopher Stanton. "Should Workplace Development Programs be Voluntary or Mandatory? Evidence from a Field Experiment on Mentorship." *NBER Working Paper*: 29148, 2021.
- ¹² Hoffman, Mitchell, and Christopher Stanton. "People, Practices, and Productivity: A Review of New Advances in Personnel Economics." *NBER Working Paper*: 32849, 2024.
- ¹³ Chandrasekhar, Arun G., Benjamin Golub, and He Yang. "Signaling, Shame, and Silence in Social Learning." *NBER Working Paper*: 25169, 2018.
- ¹⁴ Heursen, Lea, Svenja Friess, and Marina Chugunova. "Reputational Concerns and Advice-Seeking at Work." No. 447. Discussion Paper, 2023.
- ¹⁵ Emanuel, Natalia, Emma Harrington, and Amanda Pallais. "The Power of Proximity to Coworkers: Training for Tomorrow or Productivity Today?" *NBER Working Paper*: 31880, 2023.
- ¹⁶ Roghanizad, M. Mahdi, and Vanessa K. Bohns. "Should I Ask Over Zoom, Phone, Email or In-Person? Communication Channel and Predicted Versus Actual Compliance." *Social Psychology and Personality Science* 13, no. 7 (2022): 1163-1172.
- ¹⁷ Cornelissen, Thomas, Christian Dustmann, and UTA Schönberg. "Knowledge Spillovers, Competition, and Individual Careers." Mimeo, 2023.
- ¹⁸ Lazear, Edward P. "Pay Equality and Industrial Politics." *Journal of Political Economy* 97, no. 3 (1989): 561-580.